

# Maximizing Player Satisfaction: Collaborative Strategies in Two-Player Gnomes Game

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**Abstract**— The objective of this research is to develop an algorithm to synthesize equilibrium strategies in two-player semi-cooperative games, considering where both players aim to maximally satisfy their (possibly incomplete) preferences over temporal goals. The designed algorithm will find a mutually agreeable way to balance each player’s preferences in a two-player game. This approach aims to provide insights into the complexity of decision-making in semi-cooperative environments. Furthermore, preliminary research will be conducted to understand how these algorithms can be adapted to real-world scenarios, where preferences and constraints are interdependent. The algorithm will be demonstrated using a modified version of the Gnomes game.

## I. INTRODUCTION

As the world becomes more connected, it becomes important for humanity to accommodate diverse preferences in various domains. Autonomous vehicles, for instance, involve multiple stakeholders with varying priorities that must be integrated into the design to maximize overall satisfaction. Users prioritize safety and efficiency, while legislators mandate compliance with traffic laws and speed regulations. Manufacturers, on the other hand, focus on durability and fuel efficiency. These preferences, however, do not hold equal weight; legal compliance outweighs fuel efficiency, and durability might be on par with safety. This raises a critical question: how can we effectively combine these preferences, especially when players involved must make sequential decisions over an extended time horizon?

In this research, we address this problem by modeling it as a two-player game, where each player has distinct goals and preferences concerning their objectives and those of the other player. In the Gnomes Game, each player is provided with a board featuring unique wall configurations that restrict their movement during their turn. Players alternate turns, moving a shared token. When a player moves the token on their board, the token’s position is simultaneously updated on the other player’s board. (Fig. (1)) Cooperation is essential for achieving their respective goals.

For example, if Player 1’s goal lies to the west of the token but a wall obstructs Player 1’s westward movement, while Player 2’s board has no such obstruction, Player 1 would instruct Player 2 to move the token west on Player 2’s turn.

Our approach allows us to explore the dynamics of preference integration in a structured manner, providing insights that can be applied to the broader context of sequential decision-making for autonomous systems.

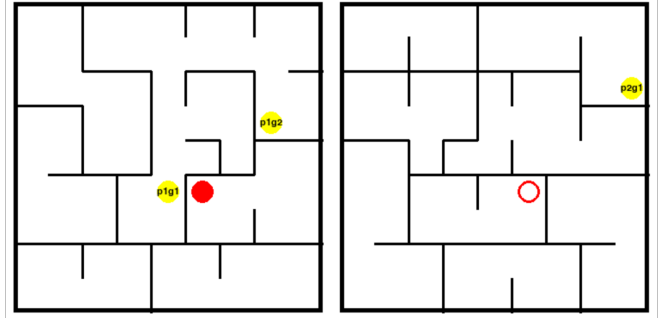


Fig. 1: Gnome Games

## II. METHODS

We employed a Pareto accumulation algorithm, which we developed based on the Pareto preference equation [2]. This equation evaluates the preferences of each stakeholder under the assumption of equal importance among stakeholders. It compares these preferences such that any deviation by a stakeholder from the algorithm’s outcome would diminish the potential satisfaction of the other stakeholder. For instance, a strategy qualifies as a Pareto equation if:

$$\begin{aligned} \forall \pi'_1 : \text{Outcome}(\pi'_1, \pi_2) \succeq_1 \text{Outcome}(\pi_1, \pi_2) \\ \implies \text{Outcome}(\pi_1, \pi_2) \succeq_2 \text{Outcome}(\pi'_1, \pi_2) \end{aligned}$$

This equation illustrates that if stakeholder one adopts an alternative strategy at any given state, it would negatively impact the potential outcome for stakeholder two at that same state. Utilizing this principle, we designed our algorithm to consider the preferences of all players and integrate them in a manner that maximizes overall satisfaction.

To achieve this, we initially constructed semi-automaton and preference graphs for both players [3], [4]. Player 1’s preferences are represented as  $(\Diamond p_1 g_1 \triangleright \Diamond p_2 g_1)$  and  $(\Diamond p_1 g_2 \triangleright \Diamond p_2 g_1)$ , while Player 2’s preferences are  $(\Diamond p_2 g_1 \triangleright \Diamond p_1 g_1)$  and  $(\Diamond p_2 g_1 \triangleright \Diamond p_1 g_2)$ . We created a semi-automaton for Player 1. (Fig. (2)) To interpret the semi-automaton graph, start from the top node, which represents the state Player 1 least prefers. As you move down through the edges connecting the nodes, you encounter progressively more preferred states, with the final state being the most preferred. For Player 1, the least preferred state is the purple node. Referring to Player 1’s preference graph (Fig. (3)), this purple node corresponds

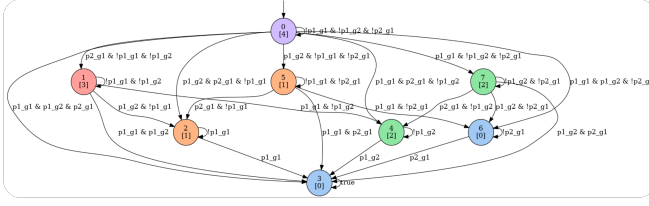


Fig. 2: Player 1's Semi-Automaton

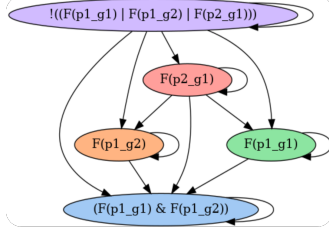


Fig. 3: Player 1's Preference Graph

to the state where none of the goals are met. In the semi-automaton, if the token reaches Player 2's goal, you transition from the purple node to the red node. According to Player 1's preference, the red node is the second least preferred state. Player 1 would prefer a move that leads to either the orange or green state. It is important to note that in the preference graph, there is no edge between the orange and green states. This indicates that these states are incomparable, meaning Player 1 has no explicit preference for one over the other but would prefer to reach either of their own goals rather than Player 2's goal. As illustrated in the semi-automaton, Player 1's most preferred state is the blue state, where both of Player 1's goals are achieved.

Similar to Player 1, Player 2's semi-automaton (Fig. (4)) indicates that the least preferred state is one where none of the goals are met. Additionally, Player 2 does not have an explicit preference between reaching Player 1's first or second goal. (Fig. (5)) However, unlike Player 1, Player 2 would prefer achieving their own goal over reaching either of Player 1's goals.

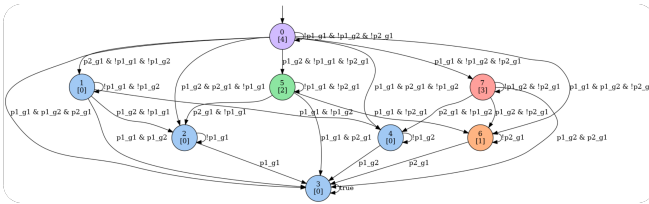


Fig. 4: Player 2's Semi-Automaton

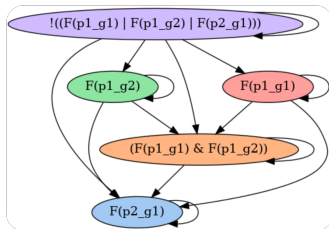


Fig. 5: Player 2's Preference Graph

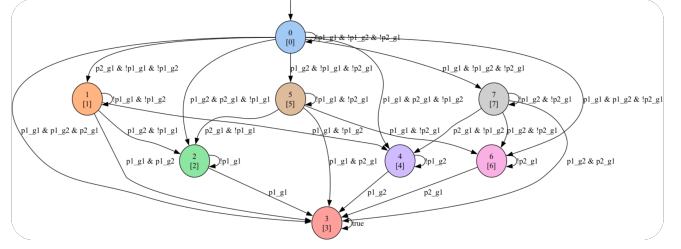


Fig. 6: Pareto Semi-automaton

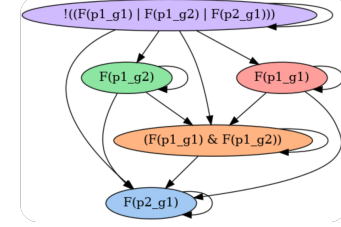


Fig. 7: Pareto Preference Graph

Using our Pareto accumulation algorithm, we integrated the semi-automaton and preference graphs of each player to construct the Pareto semi-automaton graph (Fig. (6)) and a preference graph (Fig. (7)). This combined graph, along with our Gnomes game model [1], allowed us to identify the maximum reachable states from each initial state. From this analysis, we determined the most preferred state, defined as the state with no outgoing edges leading to other nodes. Which in this case is the state where all goals are met. By backtracking through the reachable states, we identified the optimal pathway to this preferred state. This pathway represents the sequence of moves that maximizes the satisfaction of both players. With this pathway established, we devised strategies for each player. These strategies ensure that, by following the prescribed moves, each player achieves maximum satisfaction. The process involved a detailed examination of the interactions between the players' preferences and the constraints imposed by the game, ultimately leading to a cooperative solution that aligns with the principles of the Pareto equation.

### III. RESULTS

After developing the strategies, we implemented them into a simulator of the Gnomes game to verify their effectiveness in achieving each goal and reaching the preferred state. Subsequently, we applied constraints on the number of turns to evaluate the results and compared the outcomes to those achieved by a human player to assess their optimality. Under a constraint of only 10 turns, the strategies did not yield ideal results. The human player made more logical decisions, indicating superior performance. We attribute this discrepancy to the implementation details of the code. Adjustments to the code may enhance the accuracy of the strategies, potentially aligning them more closely with the logical decision-making observed in human players.

### IV. CONCLUSION

In an interconnected world, accommodating diverse preferences is crucial. This research developed an algorithm to create equilibrium strategies in two-player semi-cooperative

games, illustrated through the modified Gnomes Game. By integrating semi-automaton and preference graphs, our Pareto accumulation algorithm balances each player's preferences to ensure mutually agreeable outcomes. The study highlights the complexities of decision-making in semi-cooperative environments, showing practical applications in areas like autonomous vehicle design, where stakeholder preferences must be harmonized. Testing revealed that human players sometimes outperformed the algorithm under constraints, indicating areas for refinement. This research enhances our understanding of semi-cooperative game dynamics and provides a framework for addressing similar challenges in various domains.

## V. FUTURE WORK

Future work will expand the Pareto accumulation algorithm to multi-player games, examining the complexity and group dynamics involved. We will also explore movement constraints by limiting players to visit each state only once. Additionally, the study will investigate strategic deception, such as motivations for lying, akin to scenarios in games like Werewolf. Applying Pareto principles to real-world applications in economics, politics, and autonomous vehicle strategies will further validate the algorithm's versatility and practicality.

## VI. ACKNOWLEDGMENTS

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## REFERENCES

- [1] Shenghui Chen, Daniel Fried, and Ufuk Topcu. Human-agent cooperation in games under incomplete information through natural language communication. *arXiv preprint arXiv:2405.14173*, 2024.
- [2] A Jorge, Sheila A McIlraith, et al. Planning with preferences. *AI Magazine*, 29(4):25–25, 2008.
- [3] Abhishek Ninad Kulkarni and Jie Fu. Opportunistic qualitative planning in stochastic systems with preferences over temporal logic objectives. *arXiv preprint arXiv:2203.13803*, 2022.
- [4] Hazhar Rahmani, Abhishek N Kulkarni, and Jie Fu. Preference-based planning in stochastic environments: From partially-ordered temporal goals to most preferred policies. *arXiv preprint arXiv:2403.18212*, 2024.